Step 1. Data summary

Total students = 150

GPA (X)	Placed (P)	Not Placed (N)	Row total
High (H)	40	10	50
Medium (M)	30	20	50
Low (L)	10	40	50
Column total	80	70	150

(a) Joint & Marginal Probabilities

$$P(X=x, Y=y) = \frac{\text{count}(x,y)}{150}$$

- $P(H,P) = 40/150 = 0.2667$
- $P(H,N) = 10/150 = 0.0667$
- $P(M,P) = 30/150 = 0.2000$
- $P(M,N) = 20/150 = 0.1333$
- $P(L,P) = 10/150 = 0.0667$
- $P(L,N) = 40/150 = 0.2667$

✅ Check: sum = 1.0

Marginals:

- $P(X=H) = 50/150 = 0.3333$
- $P(X=M) = 0.3333$
- $P(X=L) = 0.3333$

ITA Classroom Activity 5

Date: 01.08.2025 (Friday)

Hour: 1 Hour

Mutual Information as a Tool for Analyzing Student Placement Outcomes

You are part of the Placement Analytics Team at your college. The institution wants to understand how **academic performance (GPA)** impacts **placement outcomes**, and whether GPA is a reliable indicator for identifying students at risk of not getting placed.

The anonymized data of 150 final-year students is categorized as follows:

GPA Category (X):

- High (H): GPA ≥ 8.0
- Medium (M): 6.5 ≤ GPA < 8.0
- Low (L): GPA < 6.5

Placement Status (Y):

- Placed (P)
- Not Placed (N)

GPA Category	Placed (P)	Not Placed (N)
High (H)	40	10
Medium (M)	30	20
Low (L)	10	40

a) Compute Joint and Marginal Probabilities.
b) Compute entropy of placement status.
c) Compute conditional entropy of placement given GPA H(Y|X).
d) Calculate Mutual Information. Based on your result:
   i. Would you recommend using GPA alone to identify students at risk of not getting placed?
   ii. What other factors (at least 2) might improve placement prediction?

- $P(X=M) = 0.3333$
- $P(X=L) = 0.3333$
- $P(Y=P) = 80/150 = 0.5333$
- $P(Y=N) = 70/150 = 0.4667$

(b) Entropy of placement status $H(Y)$

$$H(Y) = -\sum_y P(y)\log_2 P(y)$$

- $P(P) = 0.5333,\ P(N) = 0.4667$

$$H(Y) = -[0.5333\log_2(0.5333) + 0.4667\log_2(0.4667)]$$

- $0.5333\log_2 0.5333 \approx -0.483$
- $0.4667\log_2 0.4667 \approx -0.513$

$$H(Y) \approx 0.996 \text{ bits}$$

(c) Conditional entropy $H(Y|X)$

$$H(Y|X) = \sum_x P(x)H(Y|X=x)$$

Each GPA group has 50 students → $P(x) = 1/3$.

- **High (H):**
  P(P|H)=40/50=0.8; P(N|H)=0

Each GPA group has 50 students → $P(x) = 1/3$.

- **High (H):**
  $P(P|H) = 40/50 = 0.8,\ P(N|H) = 0.2$
  $H(Y|H) = -(0.8\log_2 0.8 + 0.2\log_2 0.2)$
  $= -(0.8 \times -0.322 + 0.2 \times -2.322) = 0.722$
- **Medium (M):**
  $P(P|M) = 0.6,\ P(N|M) = 0.4$
  $H(Y|M) = -(0.6\log_2 0.6 + 0.4\log_2 0.4)$
  $= 0.971$
- **Low (L):**
  $P(P|L) = 0.2,\ P(N|L) = 0.8$
  Same as High (symmetry): 0.722

Now weight:

$$H(Y|X) = \tfrac{1}{3}(0.722 + 0.971 + 0.722) = 0.805 \text{ bits}$$

(d) Mutual Information $I(X;Y)$

$$I(X;Y) = H(Y) - H(Y|X)$$

$$I(X;Y) = 0.996 - 0.805 = 0.191 \text{ bits}$$

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Interpretation

i. **Is GPA alone a good predictor?**

- The mutual information is very small (~0.19 bits out of max ~1 bit).
- This means GPA gives only **limited information** about placement.
- So **No**, GPA alone is not reliable to identify students at risk.

ii. **Other factors that may help predict placement:**

1. **Communication skills / interview performance** (often key in placement).
2. **Technical skills / coding test performance** (many companies use this).
3. (Optional extras: internships, extracurriculars, prior projects, domain specialization).

✅ Final Results:

- Joint probabilities: listed above
- Marginal probabilities: listed above
- $H(Y) \approx 0.996$ bits
- $H(Y|X) \approx 0.805$ bits
- $I(X;Y) \approx 0.191$ bits
- GPA is **not sufficient alone**, need more factors